

A Novel Low-Complexity Clock Recovery for Non-Integer Oversampling Coherent Optical Transmission

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Abstract *The non-integer oversampling technique has been widely adopted in high-speed coherent optical transmission. We propose a novel low-complexity clock recovery algorithm for recovering the clock of non-integer oversampled signals. Compared to conventional methods, the novel clock recovery algorithm reduces the number of real multiplications by 93%. ©2026 The Author(s)*

Introduction

To accommodate the increasing demand for data rates, the non-integer oversampling technique has been widely adopted in coherent optical transmission[1–4]. This technique allows the analog-to-digital converter (ADC) sampling rate to be lower than twice the baud rate [5–7]. However, in a non-integer oversampling receiver, the ADC bandwidth is lower than the signal bandwidth, leading to inter-symbol interference (ISI). Moreover, the clock recovery (CR) algorithm (e.g., Gardner's algorithm [8]) and the T/2-spaced equalizer require 2 samples per symbol (2-SPS). Therefore, in a conventional receiver DSP flow, the non-integer oversampled signal is resampled to 2-SPS first, and then the CR (Gardner's algorithm or Godard's algorithm [9]) algorithm recovers the clock of the signal. The T/2-spaced equalizer is used to compensate for ISI. However, as the oversampling ratio approaches 1-SPS, the low-pass filter requires more taps in the resampling process, leading to increased complexity. Moreover, the interpolation filter in CR also contributes to high complexity. This growing complexity becomes a bottleneck for scaling to higher baud rates. To address this issue, a modified Godard algorithm is proposed for clock recovery from non-integer oversampled signals [1, 4]. In the time domain, ref. [2] proposes a modified Lee algorithm working at 5/4-SPS.

In this paper, we propose a novel clock recovery algorithm based on a cascaded comb integrator(CCI) interpolation filter. The CCI interpolation filter consists of adders and delays, and no multiplier is required [10]. For convenience, the new clock recovery algorithm is referred to as CCICR. CCICR works under a non-integer oversampled signal and outputs a clock-recovered 2-SPS signal.

To evaluate the performance and computational complexity of CCICR, we used a dual-polarization 16QAM (DP-16QAM) signal transmitted over 100 km of standard single-mode fiber(SSMF) in the experiment, with an oversampling ratio of 5/4-SPS at the receiver. The results show that, compared with traditional CR(TCR), CCICR incurs about 0.15 dB Q^2 penalty but reduces the number of real multiplications by 93%.

Working Principle

Fig. 1 shows the diagram of CCICR. The D represents delay, and R denotes R-fold upsampling. Before inserting R-1 zeros between two adjacent original samples, the signal passes through a comb filter consisting of N cascaded delays. After inserting R-1 zeros, the signal passes through an integrator with N cascaded delays. This processing can be represented by

$$X = CCI_{\text{interp}}(X_{\text{ADC}}, R, N) \quad (1)$$

Where X_{ADC} is the signal sampled by the ADC, the X is the signal processed by the CCI filter. After the signal passes through the CCI filter, the index calculator is executed. The index calculation method can be represented by

$$i(n+1) = i(n) + \frac{RK_1}{K_2} + DM \quad (2)$$

$$DM = \begin{cases} i_c(k) & \text{mod}(n, L) = 0 \\ 0 & \text{mod}(n, L) \neq 0 \end{cases} \quad (3)$$

$$i_c(k+1) = \begin{cases} i_c(k) + 1 & PE(k) \leq -\lambda \\ 0 & -\lambda < PE(k) < \lambda \\ i_c(k) - 1 & PE(k) \geq \lambda \end{cases} \quad (4)$$

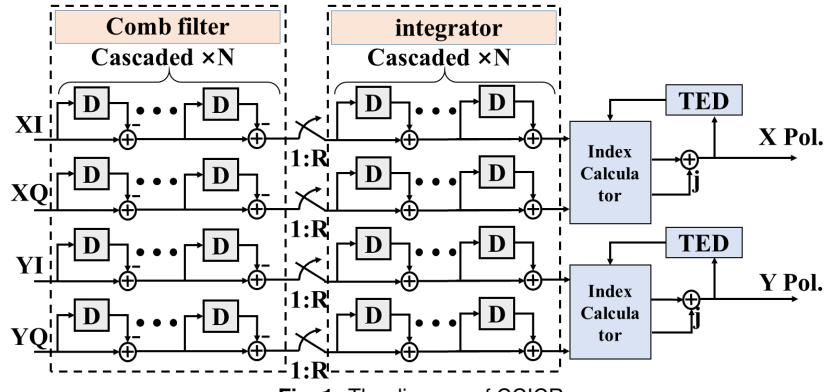


Fig. 1: The diagram of CCICR

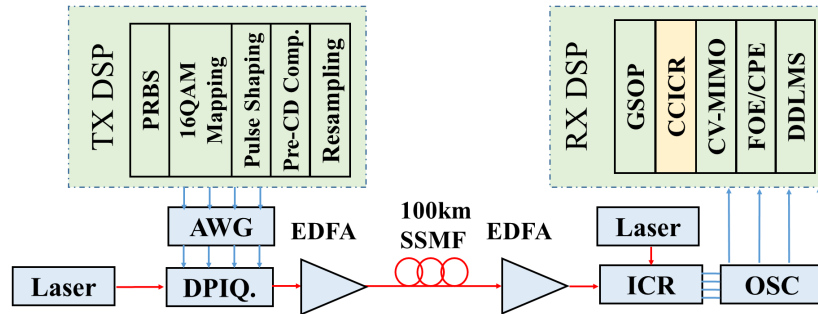


Fig. 2: The experimental setup and DSP flow, EDFA: Erbium-Doped Fiber Amplifier

where $i(n)$ is an index of X , and $X(i(n))$ is the even-indexed samples of the 2-SPS output signal, L and k is positive integers. λ is the comparison threshold. K_1, K_2 is K_1/K_2 -SPS. PE is calculated by timing error detect(TED), which can be represented by

$$PE(k+1) = \sum_{n=L(k-1)+1}^{Lk} \left\{ \sum_{m=Re, Im} \left[X_m(i(n)) - \frac{1}{2} \left(X_m(i(n) - \frac{RK_1}{K_2/2}) + X_m(i(n) + \frac{RK_1}{K_2/2}) \right) \right] \right. \\ \left. \left[\left(X_m(i(n) - \frac{RK_1}{K_2/2}) - X_m(i(n) + \frac{RK_1}{K_2/2}) \right) \right] \right\} / L \quad (5)$$

Where Re and Im represent the real and imaginary parts.

Experimental result

We investigate the performance of the proposed scheme through experiments. The experimental setup is shown in Fig. 2. Two DFB lasers, with linewidths of <100 kHz and wavelengths of 1551.12 nm, are used as the signal carrier and local oscillator, respectively. A 93.4 GSa/s arbitrary waveform generator (AWG, Keysight M8196A) is used to generate the four tributaries of the 32 GBaud DP-16QAM signal. Then the signal is modulated onto the optical carrier using a dual-polarization optical IQ modulator (DPIQ, FTM7977) with a 3-dB bandwidth of 23 GHz. The optical signal is amplified by EDFAs and transmitted over a 100-km SSMF. At the receiver, the optical signal is demodulated by an integrated coher-

ent receiver (ICR, CPRV1225A) with a 3-dB bandwidth of 29 GHz. Finally, the signal is sampled by an oscilloscope (OSC, Keysight DSOZ594A) at a sample rate of 40 GHz (5/4-SPS). The chromatic dispersion is pre-compensated in the TX DSP. In a point-to-point direct fiber link, since the fiber length is fixed, chromatic dispersion (CD) could in principle be pre-compensated within the TX DSP. If a 1310 nm laser is used for short-distance transmission, CD pre-compensation is not required in principle. In RX DSP, the CV-MIMO uses a 15-tap filter, and the DDLMS uses a 7-tap filter. In the CCICR, the number of cascaded layer $N=3$.

The CCI filter's sidelobe gain decreases as N increases. As shown in Fig. 3(a), when $N=1$, the CCI filter sidelobe gain is -13.28 dB. A high sidelobe gain leads to spectral aliasing. When $N=3$, the sidelobe gain is reduced to -39.7 dB. However, as N increases, the filter's passband gain declines more steeply with frequency. This leads to distortion(CCI distortion). Fig.3(b) shows the cost that compensates for the distortion. The TCR is a CR algorithm based on Gardner's algorithm. The BER and Q^2 factor are used to evaluate CCICR performance. The number of equalizer taps varies from 9 to 25. When using a 15-tap CV-MIMO, the Q^2 factor is 8.36 with TCR. Under the same number of taps, the Q^2 factor is 8.39 with CCICR($N=3$). For other tap lengths, CCICR ($N=3$) introduces a 0.15 dB Q^2 penalty compared to TCR. More cascaded layers improve the performance of CCICR, but need more adders and delays. Fig.3(c) shows the CCICR performance in the $N=2$ and the $N=3$ conditions. Because the CCICR has a better

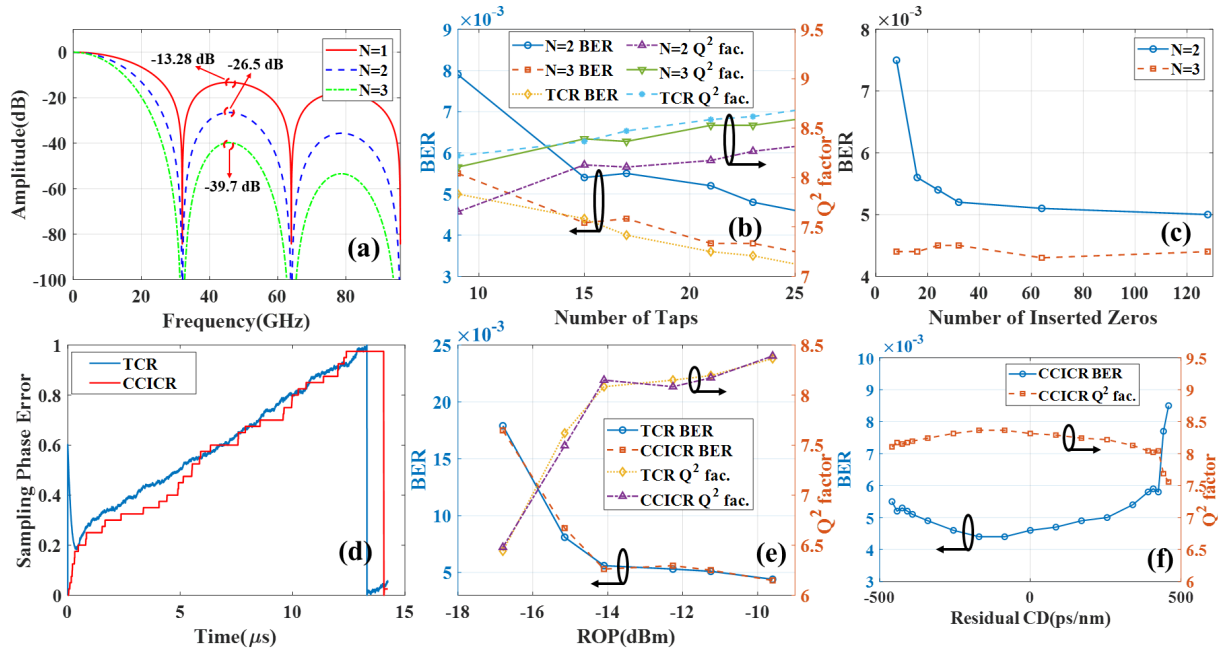


Fig. 3: (a) CCI filter frequency response under different cascaded layers, (b) Performance loss caused by CCI distortion, (c) Performance of CCICR under different cascaded layers, (d) The SPE estimated by CCICR, (e) Performance of CCICR under different ROP, (f) Residual CD influence on the system performance

low-pass characteristic when $N = 3$, the BER in $N = 3$ is better than the BER in $N = 2$. In addition, CCICR shows superior performance when the number of inserted zeros is only 7 or 15 at $N = 3$. This means CCICR does not require a large number of adders and delays. Fig. 3(d) displays the sampling phase error (SPE) estimation with CCICR. The SPE is calculated by PE in Eq. 5. Fig. 3(d) shows that the SPE estimated by CCICR has a small error compared with that estimated by TCR. In addition, the amplitude of the clock-recovered signal obtained using CCICR is different from that obtained using TCR. However, the small SPE estimation error and amplitude difference do not affect the correct demodulation of the signal. This is because the small SPE error and amplitude difference can be compensated by the subsequent T/2-spaced equalizer. Fig. 3(e) displays the CCICR performance under different received optical power (ROP) conditions. Fig. 3(e) shows that as the ROP increases, CCICR's Q^2 penalty remains consistently below 0.15 dB and even approaches zero. This means CCICR has robust performance. Finally, Fig. 3(f) shows that the CCICR can tolerate residual CD about ± 350 ps/nm.

Since multiplication is more complex than addition, the number of real multiplications required per symbol (RMPS) is used to evaluate CCICR's complexity. The Tab.1 displays the number of RMPS that CCICR and TCR are required. In Tab. 1, an N_1 -tap low-pass FIR filter is used for resampling. An N_2 -tap interpolation filter is used for CR. The value of N_1 is set to $10 \times K_2 + 1$, and the value of N_2 is set to 4. From Tab. 1, when $K_1/K_2=5/4$,

Tab. 1: Real Multiplications Required Per Symbol

	TCR	CCICR
Resampling	$2K_1/K_2 \times 2K_2/K_1 \times N_1/K_2$	0
Interpolation	$2K_1/K_2 \times 2K_2/K_1 \times N_2$	0
TED	4	4

TCR requires 41 RMPS for resampling, 16 RMPS for interpolation, and 4 RMPS for TED. Total required 61 RMPS. However, the CCICR requires 4 RMPS for resampling, interpolation and TED in total. Compared to TCR, CCICR lowers the RMPS by 93%. It should be noted that the reduction in RMPS comes at the cost of an increase in adders and delay elements.

Conclusions

In this paper, we propose a novel clock recovery algorithm for non-integer oversampling coherent optical transmission. A 5/4-SPS oversampled 16QAM signal is transmitted over 100 km of SSMF in the experiment. The performance of CCICR is studied under different numbers of cascaded layers, different tap lengths of the CV-MIMO equalizer, and different ROP. The results show that the proposed algorithm reduces the combined complexity of resampling and CR algorithm by 93%. Compared with TCR, the Q^2 factor penalty is about 0.15 dB.

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